

PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: Color Image Processing

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Color Models

Color models provide a standardized way to specify colors. They are typically categorized as hardware-oriented (RGB, YUV) or human-oriented (HSI).

1.1 RGB Color Model

The RGB model is based on a 3D Cartesian coordinate system where the primary colors Red, Green, and Blue are at three corners. The values are typically normalized to $[0, 1]$:

- Black is at the origin $(0, 0, 0)$, while White is at $(1, 1, 1)$.
- Colors are additive: combining primaries yields secondary colors (Cyan, Magenta, Yellow).

1.2 YUV Color Model

Used widely in television transmission and video compression standards (like MPEG/H.26X). It decouples brightness (Luminance) from color (Chrominance):

- **Y (Luminance)**: Represents brightness information.
- **U and V (Chrominance)**: Represent color differences.
- Conversion from RGB to YUV:

$$Y =$$

RGB to HSI Conversion Formulas

Given $R, G, B \in [0, 1]$, the HSI components are computed as:

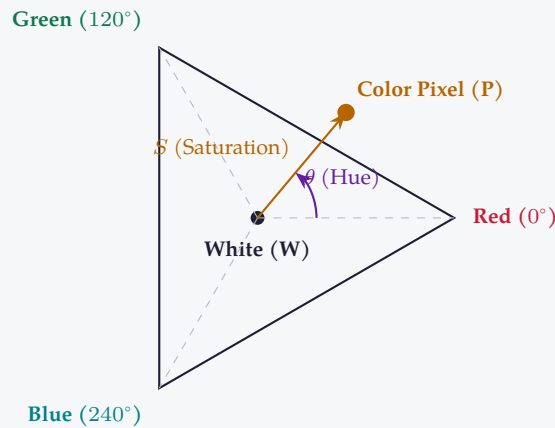
$$I = \frac{1}{3}(R + G + B) \quad (4)$$

$$S = 1 - \frac{3}{R + G + B} [\min(R, G, B)] \quad (5)$$

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360^\circ - \theta & \text{if } B > G \end{cases} \quad (6)$$

where the angle θ is:

$$\theta = \cos^{-1} \left(\frac{\frac{1}{2} [(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}} \right) \quad (7)$$

HSI Color Model Coordinates

to gray. A prototype color vector $\mathbf{a}$ and distance threshold D_0 are defined:

$$s_i = \begin{cases} r_i & \text{if } \|\mathbf{z} - \mathbf{a}\| \leq D_0 \\ 0.5 & \text{if } \|\mathbf{z} - \mathbf{a}\| > D_0 \end{cases} \quad (10)$$

- **Tone and Color Corrections:** Adjusting brightness, contrast, and gamma on color channels. Applying gamma correction on Intensity (I) in HSI space preserves the original hue and saturation, avoiding unwanted color shifts that occur when applying it directly on individual RGB channels.

3. Color Image Smoothing and Sharpening

3.1 Color Image Smoothing

Smoothing can be performed either in full-color vector space or by processing individual channels separately.

For a neighborhood S_{xy} , the average vector is:

$$\bar{\mathbf{c}}(x, y) = \frac{1}{K} \sum_{(s,t) \in S_{xy}} \mathbf{c}($$

- **Euclidean Distance Classifier:** If color features are uncorrelated, we classify pixel $\mathbf{z}$ based on Euclidean distance:

$$D(\mathbf{z}, \mathbf{a}) = \|\mathbf{z} - \mathbf{a}\| = \sqrt{(z_R - a_R)^2 + (z_G - a_G)^2 + (z_B - a_B)^2} \quad (13)$$

A pixel is segmented if $D(\mathbf{z}, \mathbf{a}) \leq D_0$.

- **Mahalanobis Distance Classifier:** Accounts for correlations between channels:

$$D_M(\mathbf{z}, \mathbf{a}) = \sqrt{(\mathbf{z} - \mathbf{a})^T \mathbf{C}^{-1} (\mathbf{z} - \mathbf{a})} \quad (14)$$

4.2 Segmentation in HSI Space

Typically performed by thresholding Hue (H) and Saturation (S) channels:

- Intensity is often ignored, making the segmentation robust

Since $B = 0.4$ and $G = 0.8$, we have $B \leq G$. Thus, Hue is:

$$H = \theta = 139.1^\circ$$

Final Result: The HSI coordinate is:

$$H = 139.1^\circ, \quad S = 0.571, \quad I = 0.467$$

Quick Revision — Module III

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Key Formulas at a Glance

Intensity (HSI)	$I = \frac{1}{3}(R + G + B)$
Saturation (HSI)	$S = 1 - \frac{3}{R+G+B} [\min(R, G, B)]$
Luminance (YUV)	$Y = 0.299R + 0.587G + 0.114B$
RGB Color Complement	$s_R = 1 - r_R, \quad s_G = 1 - r_G, \quad s_B = 1 - r_B$
RGB Euclidean distance	$D(\mathbf{z}, \mathbf{a}) = \sqrt{(z_R - a_R)^2 + (z_G - a_G)^2 + (z_B - a_B)^2}$
Mahalanobis distance	$D_M(\mathbf{z}, \mathbf{a}) = \sqrt{(\mathbf{z} - \mathbf{a})^T \mathbf{C}^{-1} (\mathbf{z} - \mathbf{a})}$

One-Line Definitions

Frequently Asked / Viva Questions (contd.)

4. What is the complement of Red in RGB space?

Under normalized RGB space, Red is $(1, 0, 0)$. Its complement is $(1 - 1, 1 - 0, 1 - 0) = (0, 1, 1)$, which represents Cyan (a secondary color).

5. What is the main advantage of segmenting color images in HSI space?

By thresholding only Hue (H) and Saturation (S) and ignoring Intensity (I), the segmentation becomes robust to shadows and spatial variations in lighting.

6. Explain the physical meaning of a Saturation value of 0.

A saturation of 0 means the color is completely diluted with white light, representing a neutral shade of gray.

7. Why do we apply gamma correction to the Intensity channel in HSI rather than RGB channels?

Applying it to RGB channels independently alters their relative ratios, causing severe shifts in hue. Correcting the Intensity channel in HSI preserves hue and saturation.